Computers Find Thousands of Hidden Steps in Turning CO2 Into Fuel

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Carbon dioxide, or CO2, is a gas released when we burn fuels. Scientists are looking for ways to turn it into useful things, such as the fuel and chemical **methanol**. Now researchers in India have used artificial intelligence to find thousands of chemical reactions that earlier models had missed.

The team works at the Indian Institute of Science, also called IISc. Their study was published in the journal Nature Communications.

One way to recycle CO2 is to make it react with hydrogen with help from a **catalyst**. A **catalyst** is a material that speeds up a chemical reaction without being used up. In this study, the **catalyst** was copper. The process involves a huge number of tiny reaction steps, too many to study one by one with powerful computer **simulations**. So scientists usually focus on only a few steps they think matter most.

The IISc team started with 152 reactions. They then trained **machine learning** programs, a type of AI, to quickly estimate how hard other reactions would be. Their network grew to 9,389 reactions.

That made a big difference. The small network wrongly predicted that the main product would be **formic acid**, not **methanol**. The bigger network matched real experiments and predicted about 40 times more CO2 being converted.

The bigger network also revealed a surprise. Hydrogen could sometimes join the reaction as a whole **molecule**, instead of first splitting into separate atoms. "The idea that hydrogen can transfer as an intact **molecule**, without first splitting into atoms, runs against what most of us were taught," said Shivam Chaturvedi, a student on the team.

The researchers say this could help scientists design better **catalysts** and study other important industrial processes.

Word Treasure

methanol -- Methanol is a simple alcohol used as a fuel and to make other chemicals.

catalyst -- A catalyst is a material that speeds up a chemical reaction without being used up.

simulations -- Simulations are computer models that copy how something works in real life.

machine learning -- Machine learning is a type of AI in which computers learn patterns from examples.

molecule -- A molecule is a tiny group of atoms joined together.

formic acid -- Formic acid is a simple chemical, also found in some ant stings.

Background you need

Carbon dioxide is a gas released when fuels such as gasoline and coal are burned, and scientists are searching for ways to recycle it into useful products.

One method combines carbon dioxide with hydrogen on the surface of a catalyst, such as copper, which helps the chemicals react.

This change happens through many tiny reaction steps, and checking each one with detailed computer simulations takes a huge amount of computing power.

Machine learning can quickly estimate how difficult each step is, letting scientists study far more reactions than before.

Break it down

LEAD: Researchers in India used AI to find thousands of overlooked reactions for turning CO2 into methanol.

BACKGROUND: CO2 reacts with hydrogen on a copper catalyst in many tiny steps

KEY EVENT: Reaction network expanded from 152 to 9,389 reactions

+- WHO: Team at the Indian Institute of Science (IISc)

+- HOW: Machine learning estimated how hard each reaction would be

EVIDENCE: Bigger network matched experiments, predicting about 40 times more CO2 converted; small one wrongly predicted formic acid

SURPRISE: Hydrogen can join as an intact molecule

OPEN QUESTION: Can this help design better catalysts and study other industrial processes?

Why it matters

Finding better ways to turn carbon dioxide into fuels and chemicals could help reduce waste gas, and the AI method could speed up research on other industrial processes.

Test what you remember

1. What did the researchers want to turn carbon dioxide into?

- (A) Useful products such as methanol
- (B) Oxygen for astronauts
- (C) Plastic bottles
- (D) Water

2. Which material served as the catalyst in this study?

- (A) Copper
- (B) Glass
- (C) Iron
- (D) Gold

3. How many reactions did the team start with?

- (A) 9,389
- (B) 10
- (C) 40
- (D) 152

4. What did the small network wrongly predict as the main product?

- (A) Hydrogen
- (B) Methanol
- (C) Formic acid
- (D) Carbon monoxide

5. How much more CO2 conversion did the bigger network predict?

- (A) About 40 times more
- (B) Ten times less
- (C) About twice as much
- (D) The same amount

6. What surprise did the bigger network reveal?

- (A) Hydrogen can join as a whole molecule without first splitting into atoms
- (B) Copper melts in the reaction
- (C) CO2 cannot be recycled
- (D) Methanol turns back into CO2

Answer key (try the quiz first!): 1: A 2: A 3: D 4: C 5: A 6: A

Questions to think about

1. Shivam Chaturvedi, IISc student: He says the finding that hydrogen can move as an intact molecule goes against what most scientists were taught.

2. The research team: The researchers say the discovery could help design better catalysts and study other important industrial processes.

MY THOUGHTS (at least 20 words):